

# TrustLayer

## *Real-Time Meeting Identity & Deepfake Verification Agent*

*Supervisor*

*Sir Farooq Javed*

*Group#02*

*BSCS51F23R019*

*BSCS51F23R009*

*BSCS51F23S006*

## 1. Introduction

Video conferencing is now the default medium for executive decisions, financial approvals, board governance, and remote hiring. Real-time face-swap and voice-cloning tools can impersonate a participant convincingly enough to fool human judgment under time pressure. Documented cases already exist — including an employee who authorized a multi-million-dollar transfer after a call in which every other participant was an AI-generated deepfake.

Current platforms only verify that a meeting link was joined — not that the person on camera is real, is who they claim to be, or is authorized to approve the request being made. TrustLayer is an AI agent that joins meetings as an automated observer, fusing three independently trained models — facial forensics, voice anti-spoofing, and liveness/behavioral analysis — into one risk-scoring agent. It alerts the host as soon as an anomaly appears and produces an auditable report after the meeting.

## 2. Purpose

- Continuously verify, during the live call, that every participant is a live human and not synthetic media.
- Give the host an immediate, actionable alert (participant, confidence score, contributing signals, recommended action) when someone is flagged.
- Provide a structured, auditable post-meeting record (attendance, verification outcomes, flagged incidents) for compliance, legal, and insurance use.
- Demonstrate that a modular, decision-level fusion design can adapt as deepfake generators evolve.

## 3. Problem Statement

Video conferencing is now where executives approve payments, boards make governance decisions, and companies hire remotely. Platforms such as Zoom, Teams, and Meet only verify that someone joined via a link — not that the person on camera is who they claim to be, is a real human, or is authorized to approve what is being asked.

Meanwhile, real-time face-swap and voice-cloning tools run on a single consumer GPU and can impersonate a known executive well enough to defeat human judgment under time pressure. Real incidents already exist, including a multi-million-dollar wire transfer approved on a call where every other participant was a deepfake.

No widely deployed tool continuously verifies participant authenticity inside a live meeting and alerts the host the moment synthetic or unauthorized presence is detected. Organizations currently depend on human intuition, which fails against high-quality synthetic media.

**Summary: Attackers can convincingly impersonate people in live meetings, and current platforms have no real-time way to detect it or warn the host.**

## 4. Objectives

- Design and train independent AI models capable of detecting facial deepfakes, synthetic/cloned voices, and non-human liveness patterns in real time.
- Fuse the outputs of these models into a single risk-scoring agent capable of contextual decision-making during a live meeting.
- Integrate the agent as an automated participant (“bot”) within mainstream video-conferencing platforms.
- Provide the meeting host with an immediate, actionable alert when a participant is flagged as synthetic, unauthorized, or anomalous.
- Automatically generate a structured, auditable post-meeting report summarizing attendance, verification outcomes, and flagged incidents.

## 5. Proposed Solution

TrustLayer joins a meeting as an automated observer and verifies in real time that every participant is a live, authenticated human. It works in five steps:

- **Ingest** — The agent joins through the Zoom, Teams, or Meet SDK (or WebRTC) and captures each participant's audio and video as time-aligned segments.
- **Detect** — Three independently trained models analyze each participant in parallel: facial forensics, voice anti-spoofing, and liveness/behavioral analysis.
  - Facial forensics detects face-swap, reenactment, and full facial synthesis, and highlights the suspicious region for explainability.
  - Voice anti-spoofing detects cloned or replayed speech and checks lip-to-audio synchrony.

- Liveness/behavioral analysis tracks blinking, gaze, and micro-expressions, and can issue active challenges (e.g. “turn your head”) for high-stakes moments.
- **Fuse** — A risk-scoring agent combines the three confidence scores at the decision level into one calibrated risk score and a recommended action (allow / flag / escalate), raising sensitivity in contexts like financial approvals. Decision-level fusion means each model can be retrained or replaced independently without redesigning the pipeline.
- **Alert** — When the score crosses a configurable threshold, the host is notified immediately via Slack, Teams, SMS, or email, naming the participant, confidence score, contributing signals, and a recommended action.
- **Report and store** — After the meeting, an LLM turns the risk timeline and transcript into a structured report covering attendance, verification status, and flagged events. Encrypted logs, flagged snippets, and reports go into access-controlled storage for compliance, legal, and insurance use.

What makes it different: the detection models are built from first principles rather than wrapping a third-party API; host alerting and the audit workflow get equal weight with detection; and the add-on design means organizations keep their existing conferencing tools.

## 6. Scope

### Included

- Design, training, and validation of the three detection models: facial forensics, voice anti-spoofing, and liveness/behavioral analysis.
- A fusion and risk-scoring agent with contextual weighting (e.g. higher sensitivity during financial approvals), producing an allow / flag / escalate decision.
- A meeting-ingestion bot for Zoom, Microsoft Teams, and Google Meet.
- Real-time host alerting via Slack, Teams, SMS, and email.
- LLM-generated post-meeting reports and encrypted evidence storage with role-based access and retention policies.
- Evaluation on held-out data and simulated meetings, including deepfakes from generators not seen in training.

### Excluded (future scope)

- Building a new video-conferencing platform — TrustLayer works as an add-on inside existing ones.
- Remote photoplethysmography (rPPG) and other advanced physiological liveness signals.
- On-device/edge deployment.
- Identity verification beyond meetings (help-desk calls, remote onboarding, KYC).

- Cross-organization trust networks and automatic model retraining.
- Automatic enforcement — TrustLayer alerts and recommends; the host decides whether to remove a participant or halt an approval.
- Guaranteeing 100% detection — performance is measured against stated targets, and cross-dataset generalization to unseen generators remains an open research problem.
- Resolving biometric-privacy law for real-world deployment (the project follows a consent/privacy process for its own data only).
- Deepfake detection in recorded or pre-recorded media — the focus is live meeting streams.

## 7. Stakeholders / Users

| Stakeholder                                         | Type                           | Role / Interaction                                          | Key Needs and Expectations                                                                 |
|-----------------------------------------------------|--------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Meeting host / organizer                            | Primary user                   | Invites TrustLayer, receives alerts, decides how to act     | Fast, clear alerts (<3s) with participant, score, and recommended action; few false alarms |
| Executives and senior leadership                    | Primary user / protected party | Attend high-stakes meetings; frequent impersonation targets | Protection of identity and decisions; minimal friction                                     |
| Finance and approval teams                          | Primary user                   | Act on payment/transfer requests made during calls          | Confidence the requester is genuine before money moves                                     |
| HR and recruiters                                   | Primary user                   | Conduct remote interviews and hiring calls                  | Detection of fake candidates and synthetic identities                                      |
| Meeting participants (employees, partners, vendors) | Monitored party                | Are analyzed during the meeting                             | Transparency, fair treatment, privacy of audio/video data                                  |

## 8. Literature Review / Related Work

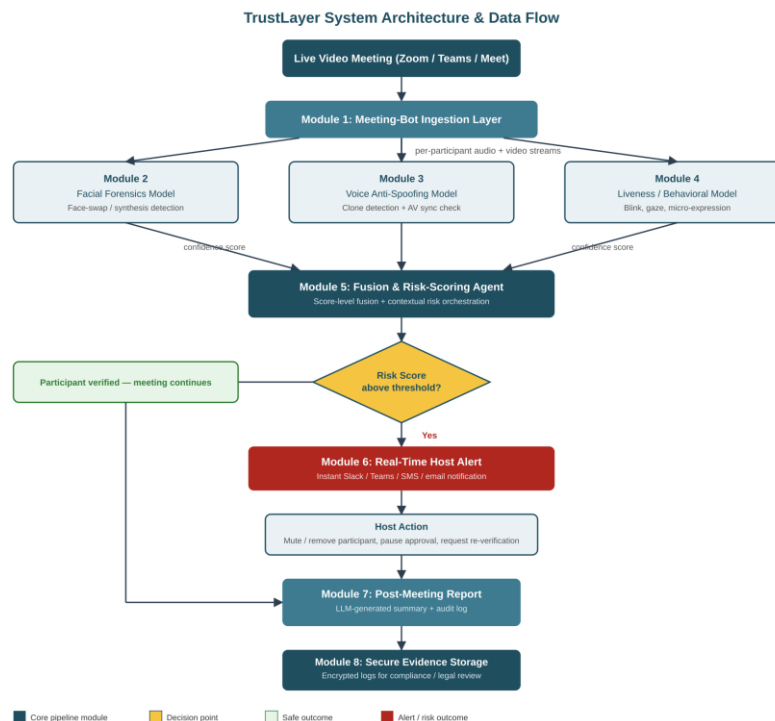
Deepfake generation has progressed through three dominant architectures — autoencoder-based face-swapping, generative adversarial networks (GANs), and, most recently, diffusion models, which now lead on visual quality and require only a handful of reference images to condition a pre-trained generator. On the detection side, academic benchmarks such as FaceForensics++, Celeb-DF, and DeeperForensics-1.0 remain the standard training corpora, and recent competitions (e.g. the NTIRE Robust AI-Generated Image Detection challenge) show that ensemble architectures combining a large self-supervised backbone with multiple specialized streams (texture, localized-facial, semantic) generalize best — though cross-dataset generalization to unseen generators remains an open research problem.

Commercially, vendors such as Pindrop, Reality Defender, Resemble AI, and GetReal Security already offer real-time meeting-deepfake detection combining voice, video, and behavioral signals, confirming both the validity of the problem and the multi-modal fusion approach adopted in this project. TrustLayer differentiates itself by building the detection models independently from first principles rather than integrating a third-party API, and by placing equal emphasis on the host-alerting and post-meeting audit workflow.

## 9. Module Breakdown

- Module 1: Meeting Ingestion Layer
- Module 2: Facial Forensics Model
- Module 3: Voice Anti-Spoofing Model
- Module 4: Liveness & Behavioral Model
- Module 5: Fusion & Risk-Scoring Agent
- Module 6: Real-Time Alerting
- Module 7: Post-Meeting Report Generator
- Module 8: Secure Evidence Storage

## 10. Workflow Diagram



## 11. Datasets

| Dataset             | Fake    | Real   | Year | Note/Description                                      | Access Link              |
|---------------------|---------|--------|------|-------------------------------------------------------|--------------------------|
| Celeb-DF (v2)       | 5,639   | 590    | 2019 | High quality                                          | <a href="#">Download</a> |
| DFDC                | 104,500 | 23,564 | 2019 | DFDC competition on Kaggle                            | <a href="#">Download</a> |
| FaceForensic++      | 5,000   | 1,000  | 2019 | Five different generation methods                     | <a href="#">Download</a> |
| DeeperForensics-1.0 | 10,000  | 50,000 | 2020 | Adds real-world perturbations                         | <a href="#">Download</a> |
| Celeb-DF++          | 53,196  | 590    | 2025 | Large-scale, challenging benchmark for generalization | <a href="#">Download</a> |

## 12. Tools and Technology Stack

| Layer                     | Technology / Tool                                                        |
|---------------------------|--------------------------------------------------------------------------|
| Meeting integration       | Zoom SDK, Microsoft Graph/Teams SDK, Google Meet API, WebRTC             |
| Facial forensics model    | Python, PyTorch, pre-trained transformers (based on requirement), OpenCV |
| Voice anti-spoofing model | Python, PyTorch/TensorFlow                                               |
| Liveness/behavioral model | MediaPipe, OpenCV                                                        |
| Fusion & orchestration    | Python, scikit-learn                                                     |
| Backend services          | FastAPI, PostgreSQL                                                      |
| Alerting                  | Slack API, Microsoft Teams webhook, Twilio (SMS), SMTP (email)           |
| Reporting                 | LLM API (e.g. Claude) for natural-language report generation             |
| Deployment                | Docker containers, cloud GPU inference (initial)                         |
| Version Control           | Git + GitHub                                                             |